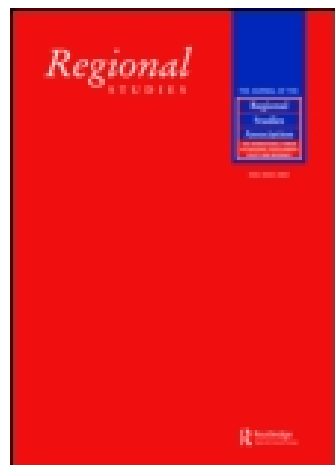




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Small Firms, Borrowed Size and the Urban-Rural Shift

N. A. Phelps^a, R. J. Fallon^b & C. L. Williams^c

^a School of Geography, University of Leeds, Leeds, LS2 9JT, UK

^b Environment and Regeneration Department, Planning Division, Luton Borough Council, Town Hall, Luton, LU1 2BQ, UK

^c CB Hillier Parker, 77 Grosvenor Street, London, W1A 2BT, UK

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Small Firms, Borrowed Size and the Urban–Rural Shift

N. A. PHELPS*, R. J. FALLON† and C. L. WILLIAMS‡

*School of Geography, University of Leeds, Leeds LS2 9JT, UK. Email: n.phelps@geog.leeds.ac.uk

†Environment and Regeneration Department, Planning Division, Luton Borough Council, Town Hall, Luton, LU1 2BQ, UK.

‡CB Hillier Parker, 77 Grosvenor Street, London W1A 2BT, UK.

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PHELPS N. A., FALLON R. J. and WILLIAMS C. L. (2001) Small firms, borrowed size and the urban–rural shift, *Reg. Studies* **35**, 613–624. In this paper we suggest that some elements of the observed urban–rural shift in firm formation and growth may centre on the phenomenon of ‘borrowed size’. In doing so, we make a link between literature on the geography of small firm formation and growth, the theory of agglomeration and the relations among settlements within urban systems. Spatial externality fields have expanded over time so that small firms can locate in smaller settlements and yet access specialized labour and informational external economies of nearby larger urban areas. We examine the relevance of aspects of borrowed size in the relationship of small firms in Leighton Linslade and Lichfield to the local economy and nearby larger urban centres. In particular we question the contribution of local institutional support to the growth of rural enterprise stressed in recent analysis. We conclude by briefly noting the implications of borrowed size for conceptions of the local economy and local economic policy.

Small firms Borrowed size External economies Urban–rural shift

PHELPS N. A., FALLON R. J. et WILLIAMS C. L. (2001) Petites entreprises, taille empruntée et déplacement urbano-rural, *Reg. Studies* **35**, 613–624. Cet article cherche à avancer la proposition suivante: il se peut que certains éléments du déplacement urbano-rural observé de la constitution et du développement des entreprises portent sur la notion de ‘taille empruntée’. Ainsi, on établit un lien entre la documentation sur la géographie de la constitution et de la croissance des petites entreprises, la théorie de la concentration géographique et les rapports entre les zones d’habitation au sein des systèmes urbains. La portée des effets externes géographiques s’est répandue sur le temps, ce qui a permis aux petites entreprises de s’implanter dans des zones plus petites et pourtant d’avoir accès à une main-d’oeuvre spécialisée et aux économies externes d’information dont jouissent les zones urbaines plus grandes qui sont à proximité. On examine la pertinence de certains éléments de la notion de taille empruntée quant au rapport des petites entreprises situées à Leighton Linsdale et à Lichfield avec l’économie locale et les grands centres urbains à proximité. En particulier on remet en question la contribution du soutien local institutionnel à la croissance des entreprises rurales, sur laquelle des analyses récentes mettent l’accent. Pour conclure, on note les retombées de la notion de taille empruntée pour ce qui est des conceptions de l’économie locale et de la politique économique locale.

Petites entreprises
Économies externes liées à la notion de taille empruntée
Déplacement urbano-rural

PHELPS N. A., FALLON R. J. und WILLIAMS C. L. (2001) Kleine Firmen, geliehene Größe und Verlegung von der Stadt aufs Land, *Reg. Studies* **35**, 613–624. In diesem Aufsatz geben die Autoren ihrer Vermutung Ausdruck, daß manche Elemente der beobachteten Verlegung von Firmen von der Stadt aufs Land bei ihrer Bildung und ihrem Wachstums auf das Phänomen der ‘geliehenen Größe’ zurückgehen. Dadurch wird eine Verbindung zwischen Literatur über die Geographie der Bildung und des Wachstums kleiner Firmen, der Theorie der Ballung und Beziehungen unter Niederlassungen in städtischen Systemen hergestellt. Räumliche Externalitätsgebiete haben sich im Laufe der Zeit ausgedehnt, so daß kleine Firmen sich in kleineren Ortschaften niederlassen können, sich jedoch gleichzeitig Zugang zu Fachkräften und nahe gelegenen externen Informationswirtschaften größerer Stadtgebiete erhalten. Es wird die Relevanz bestimmter Aspekte geliehener Größe in der Beziehung kleiner Firmen in Leighton Linslade und Lichfield zur örtlichen Wirtschaft und zu nahe gelegenen größeren städtischen Zentren untersucht. Besonderer Prüfung unterzogen wird die Gewährung örtlicher institutioneller Unterstützung des Wachstums ländlicher Unternehmen, die in kürzlich veröffentlichten Analysen hervorgehoben wurde. Den Abschluß bildet eine kurze Erwähnung der Implikationen geliehener Größe für Vorstellungen von Wirtschaft und Wirtschaftspolitik am Orte.

Kleine Firmen Geliehene Größe
Externale Wirtschaften
Verlegung von der Stadt aufs Land

INTRODUCTION

Conventional wisdom holds that there is a close relationship between small firms and localized external economies. Small firms with their limited internal resources are crucially dependent upon large and well-functioning property, labour and intermediate input markets as well as localized flows of knowledge and expertise. Historically, large cities have been thought to offer the most appropriate environment for the formation and growth of small firms (HOOVER and VERNON, 1959). Of course much has happened to the economies of large cities in advanced economies to question whether this remains the case. Nevertheless the relationship between small firms and localized external economies continues to be stressed in academic debate and policy prescriptions concerning flexible forms of production (see, for example, PIORE and SABEL, 1984; SCOTT, 1988). This literature has been influential in informing much policy prescription regarding small firms and local economic development. In the UK, an emphasis on restoring the relationship between small firms and external economies in economically troubled localities has become apparent (e.g. WILSON, 1995) with greater networking between firms being advocated alongside the build-up of, and improved co-ordination among, local institutions involved with delivering services to small firms (e.g. HAUGHTON, 1993; DEPARTMENT OF TRADE and INDUSTRY, 1994). With their lineage in Alfred Marshall's original formulation of the external economies available within individual self-contained industrial districts, these accounts and prescriptions often fail to fully recognize the relational nature of external economies within urban systems.

In important respects, then, the empirical evidence available on small firm formation and growth is at odds with these sorts of policy prescriptions. For example, Curran and Blackburn's provocatively sub-titled book, *Small Firms and Local Economic Networks: The Death of the Local Economy?*, argues strongly that there is almost a complete lack of integration of small firms with their respective local economies in anything other than a symbolic sense (CURRAN and BLACKBURN, 1994). Similarly, for some time now an urban-rural shift in small firm formation and growth has been apparent (KEEBLE, 1993; KEEBLE and TYLER, 1995). This urban-rural shift begs questions about the presumed connection between localized external economies and small firm formation and growth (see also OAKLEY and COOPER, 1989) given the growth of entrepreneurial behaviour away from established cores of population and economic activity. This shift may have much to do with the external diseconomies now characteristic of large urban areas, though KEEBLE *et al.*, 1992; KEEBLE and TYLER, 1995; and VAESSEN and KEEBLE, 1995, have produced evidence to suggest that it also derives from selective outmigration of entrepreneurs and

greater enterprising behaviour among small firms in smaller settlements and rural areas. In a recent review of the US and UK evidence, NORTH, 1998, argues that much of small and medium sized enterprise (SME) activity has tended to be concentrated in rural areas adjacent to metropolitan areas – an effect which, following ALONSO, 1973, we might term 'borrowed size'. In the UK, there is considerable evidence that businesses in small settlements are able to borrow size from Greater London as part of the regionalized economy of the South East (e.g. ALLEN, 1992; COE and TOWNSEND, 1998). There is less evidence of this effect outside the South East (KEEBLE and TYLER, 1995), although HITCHENS *et al.*'s, 1994, study of business service use by manufacturing firms in mid Wales does reveal patterns of sourcing centred upon major urban areas such as Birmingham and London.

In this paper we suggest that the concept of borrowed size is of value in understanding, in part, geographical patterns of small firm formation and growth in the UK. The concept of borrowed size suggests the need to move away from thinking of the relationship between small firms and external economies purely in terms of the characteristics of individual localities and toward an emphasis upon the *interactions* between businesses in groups of localities. Borrowed size means that different definitions of 'local' may be relevant to understanding the geographical availability of different external economies and hence appropriate points of leverage for policy. Here we consider some of the implications of this line of thought drawing upon a small empirical study of the small firms sector in Leighton Linlade and Lichfield.

THE LOCAL ECONOMY, EXTERNAL ECONOMIES AND BORROWED SIZE

As far as we are aware the concept of borrowed size has not been developed explicitly since it was first briefly described by ALONSO, 1973. According to Alonso: 'The concept of a system of cities has many facets, but one of particular interest ... is the concept of borrowed size, whereby a small city of metropolitan area exhibits some of the characteristics of a larger one if it is near other population centres' (*ibid.*, p. 200). Alonso was thinking of the tendency for people and businesses to retain the advantages of being based in smaller settlements (e.g. less congestion, lower rents) whilst also being able to reap the advantages on offer in larger settlements (e.g. access to sizeable markets, business services and expertise, larger and more diverse labour markets and cultural amenities).¹ Alonso's description of borrowed size grew out of a consideration of patterns of urbanization in the US. Arguably, however, the concept is of broader relevance to understanding the likes of networks of urban nodes in some European countries (e.g. see BATTEN, 1995), the economies of some suburban or edge urban settlements

(PHELPS, 1998) and geographical patterns of small firm formation and growth in the UK, especially the regionalized economy of the South East of England (e.g. ALLEN, 1992; COE and TOWNSEND, 1998). *Interaction* between firms and populations of individual settlements is both a measure of, and the mechanism enabling, borrowed size. With this emphasis on interaction, the concept is closely related to earlier work on the ‘non-place urban realm’ (WEBBER, 1963, 1964) and city-systems (e.g. BOURNE, 1975; PRED, 1976) and recent work on networking (see BRYSON and DANIELS, 1998, for a discussion related to small firms). However, in contrast to the tendency for ‘accessibility to be freed from propinquity’ made famous by WEBBER, 1963, p. 109), the concept of borrowed size is a manifestation of the complementarity between the advantages firms draw from their locality with those they draw from other nearby settlements through their business transactions and interactions. To an extent, then, the concept of borrowed size is capable of capturing some of the coexisting tendencies of agglomeration and dispersal of economic activity and population (see CHAMPION, 1988; CHESHIRE, 1995).

The businesses of smaller settlements are able to borrow size in this way because of the enlargement of spatial externality fields over time (PHELPS, 1992). A distinction is usually made between pecuniary (or market size related) and technological externalities (e.g. SKITOVSKY, 1954; BROWN and JACKSON, 1985; HARRISON, 1992).² This distinction actually cuts across the familiar Marshallian trinity of material input–output transactions, labour market pooling and technological externalities. The first and last of this trinity clearly embody pecuniary and technological externalities respectively but labour market pooling is reflected both in firms’ cost structures and their innovative potential.

Major urban areas or heavily urbanized regions may still offer significant markets and pecuniary external economies for some sectors. However, given improvements in transport technology and infrastructure and the growing ‘roundaboutness’ of industry, the need to locate in or near major settlements as these represent major markets has long-since been diminishing for a growing number of manufacturing and service industries (see BAIROCH, 1988, p. 344). Thus, in general terms, the localized material input–output linkages which constitute part of the trinity of Marshallian externalities now play a less critical role in promoting the agglomeration of firms (PHELPS, 1992; GLAESER, 1998; MASKELL *et al.*, 1998). The ‘new economic geography’ suggests that these pecuniary externalities – especially those associated with material input–output transactions – tend to operate over very broad geographical areas (MARTIN, 1999), such as whole continents, so that peripherality is unlikely to greatly hinder access to markets and pecuniary externalities for a wide range of industries – especially manufacturing

industries. Technological external economies, however, remain a key ingredient in highly localized forms of agglomeration. The spatial externality fields associated with labour market pooling are more circumscribed and still tend to present some constraints on firm location – though increased personal mobility means that the pecuniary and technological externalities associated with labour markets can promote relatively spatially diffuse forms of agglomeration as in the case of the British motor sport industry (PINCH and HENRY, 1999). It is technological externalities, and more particularly those embodied in the inter-personal contacts among highly skilled workers, that remain the most geographically constrained and continue to constitute the most powerful force to agglomeration (GLAESER, 1998; MASKELL *et al.*, 1998) as has been the case with head-office functions (THORNGREN, 1970; TORNQVIST, 1970; GODDARD, 1975) and some high-technology industries where ‘untraded interdependencies’ are important (STORPER, 1995). Whilst in some instances, notably globally competitive industry clusters (e.g. ‘Silicon Valley’), specialized communities of business interests are highly localized, WEBBER’s, 1963, notion of the non-place urban realm is useful in stressing the fact such ‘networking’ can also involve important non-local ties such that a degree of decentralization is possible.

Because the phenomenon of borrowed size derives from the tension between agglomeration based on technological externalities, on the one hand, and decentralization based on the more widely available pecuniary externalities, on the other, it is most likely to figure in the locational patterns of industries for which important pecuniary externalities and technological externalities are available at the scale of major conurbations or even heavily urbanized regions. There are a relatively wide range of industries in which technological external economies *can* be important including relatively low-technology manufacturing industries (e.g. see MASKELL *et al.*, 1998). However, there is a narrower range of industries for which pecuniary externalities are available at the scale of major cities or city regions – in other words for which urbanization economies are still important. These would include business services which are linked to important markets offered by the financial and head-office activities of global cities but also, for example, the likes of communications, health services and specialized consumer services for whom urbanization economies appear to be important (O’HULLACHAIN and SATTERTHWAIT, 1992). For these sectors, major urban areas are sources of *both* pecuniary and technological externalities. Global city regions with their concentrations of financial services and corporate head offices are likely to be important sources of both pecuniary and technological external economies for producer service industries so that firms in smaller accessible rural settlements are tied through technological externalities to central city

economic activities but are able to borrow the benefits of urban size without being physically located within urban cores.³ The importance of technological externalities to borrowed size appears implicit when KEEBLE and TYLER, 1995, speak of successful firms outside of large urban areas having owners/managers with urban derived knowledge. So, for example, it has been suggested that such personal contacts comprise *the* regulating institution of contemporary forms of urban agglomeration (DURANTON, 1999). Here it seems likely that borrowed size will be an effect which is geographically skewed within a city system with a high degree of primacy like the UK. For example, an ability to dip into networks of information and expertise in major settlements seems likely to be an important feature of small firm formation and growth in smaller settlements in the South East (COE and TOWNSEND, 1998) but possibly rather less important further away from London, where an over-reliance on purely local sources of advice and information may inhibit small firm performance (BRYSON and DANIELS, 1998).

We now turn to an exploration of these ideas in relation to the small firm sector in two case study settlements.

RESEARCH METHODOLOGY

Our interest in this paper is in exploring the relevance of borrowed size to the locational dynamics of small firms in accessible rural areas. Given that borrowed size is likely to figure in the location patterns of a relatively narrow range of nevertheless important service industries, we chose to focus on small firms in the business services sector.⁴ The empirical material presented here is small scale and exploratory in nature – originating with a single case study of small firms in one accessible rural settlement within the South East region but later supplemented with a second case study outside the South East.

The main element of the study consisted of separate telephone surveys of part of the small firm sector in the two settlements. Leighton Linslade and Lichfield were chosen as suitable but contrasting case study settlements in which to explore the possible relevance of borrowed size to the dynamics of small business service firms.⁵ Both settlements were selected specifically as those whose collections of firms might be able to borrow size given their proximity to nearby major conurbations (see Fig. 1). The case study nature of the research necessitated that these quite sizeable settlements were examined in order that sufficient numbers of small business service firms could be surveyed. With populations of 31,855 and 27,577 respectively in 1991, Leighton Linslade and Lichfield are much larger than, for example, KEEBLE and TYLER's, 1995, definition of rural settlements (less than 10,000 population). As such, there are clear limitations with our commentary on the economic basis of accessible rural settlements

and informal comparisons with much of the extant literature on this subject.

Leighton Linslade is composed of two smaller settlements – Leighton Buzzard and Linslade – which have merged through rapid recent population growth of which at least some is associated with employment expansion. In contrast, Lichfield, as a city, is a settlement with a much longer history and single identity. Lichfield was selected as a settlement in which to examine whether borrowed size was an important element of small firm start-up and growth in an accessible rural area outside of the South East of England. A number of factors have contributed to the recent favourable performance of the Lichfield economy including 'the general urban–rural shift in manufacturing employment and the continuing decentralisation of firms from the conurbation to lower cost peripheral locations which have reasonable access to the motorway network' (ECOTEC RESEARCH AND CONSULTING LTD, 1995, p. 20). Both case study settlements are close to major conurbations and other large free standing towns or cities (see Fig. 1) and offered the opportunity to examine the networks of interaction in which small firms are embedded and through which borrowed size might be manifested.

Sample frames of all small firms and establishments in the business services sector in Leighton Linslade and Lichfield were drawn up using local and national business directories and the *Yellow Pages*. Small firms and establishments were defined as those with 100 or fewer employees. Initially 71 firms and establishments were identified in Leighton Linslade and 82 in Lichfield. Firms were invited to answer a number of questions over the telephone, though in the event a proportion of firms requested that questionnaires be sent and returned by mail. This was a particular problem in the Lichfield study which in reality consisted in roughly equal measure of telephone responses and faxed or mailed questionnaire returns. The choice of a telephone survey was dictated by the fact that, whilst some aspects of borrowed size as it may affect the location of small firms might be readily captured quantitatively, an appreciation of its relevance to small firms might also entail a qualitative element, given the limited resources available for the study. The questionnaire was designed to elicit information regarding the geographical locus of different external economies. For instance, sources of labour, material and technical inputs as well as the perceived advantages and disadvantages of the two settlements as business locations were examined.

Of the firms initially contacted, a large number had closed or relocated or were found not to fall within the industry definitions adopted. This problem was particularly acute for Lichfield where there appeared to be significant local movement out of Lichfield city to industrial estates elsewhere in the local authority area. This left valid sample frames of 65 firms in

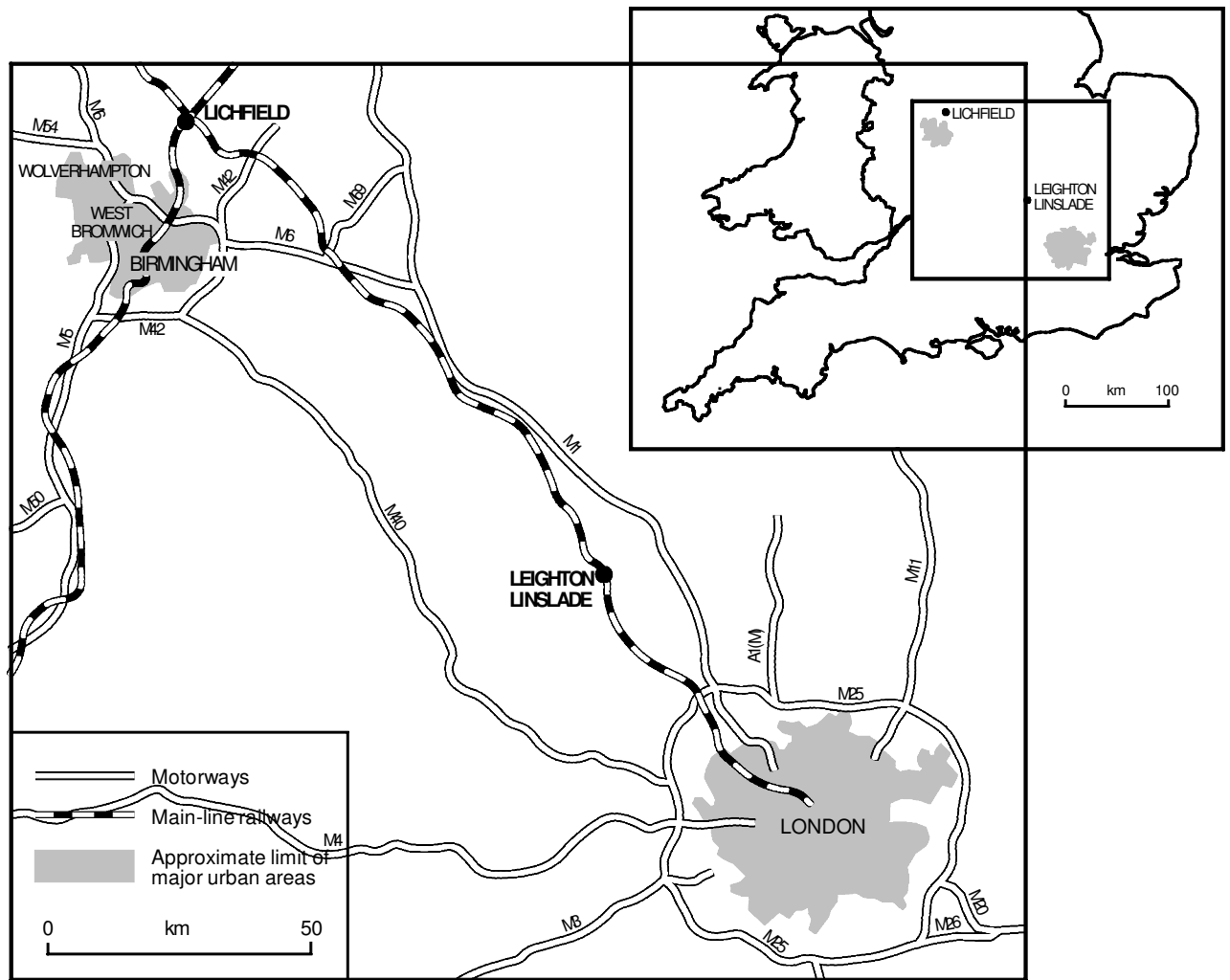


Fig. 1. Location of Leighton Linlade and Lichfield

Table 1. Details of the sample frame and responses

	Leighton Linlade		Lichfield		Total	
	Sent	Received	Sent	Received	Sent	Received
8370 Professional and technical services	10 (15.4)	9 (14.1)	9 (17.0)	7 (20.0)	19 (16.1)	16 (16.0)
8380 Advertising	2 (3.1)	2 (3.1)	4 (7.5)	2 (5.7)	6 (5.1)	4 (4.0)
8394 Computer services	30 (46.2)	30 (46.9)	11 (20.8)	7 (20.0)	41 (34.7)	37 (37.0)
8395 Miscellaneous business services	23 (35.4)	23 (35.9)	29 (54.7)	20 (55.5)	52 (44.1)	43 (43.0)
Total	65 (100.0)	64 (98.5)	53 (100.0)	36 (67.4)	118 (100.0)	100 (100.0)

Note: Percentage figures in parentheses.

Leighton Linlade and 53 in Lichfield (see Table 1). Completed questionnaires were obtained for 64 (98.5%) firms in the initial study of Leighton Linlade and 36 (67.4%) in the second study centred on Lichfield. The second, follow up study suffered particularly from a number of outright refusals to take part in the survey as well as non-return of questionnaires which firms requested be sent by mail or fax. A chi-squared test ($\chi = 3.54$) suggested that there were no significant

differences in the response rates of firms surveyed in the two case study settlements though the result here was marginal. Table 1 also shows the different composition of the business services sectors in the two case study settlements, with Leighton Linlade having a larger number of computer-related consultancy firms and Lichfield having a large number of more general business services firms (especially personnel recruitment agencies).

SMALL FIRMS IN LEIGHTON LINSLADE AND LICHFIELD: SOME ASPECTS OF BORROWED SIZE

The following discussion is divided into four sections. Firstly, the characteristics of small firms surveyed in Leighton Linslade and Lichfield are briefly recounted. Next the pecuniary externality fields of surveyed firms are explored by looking at the geography of input and output linkages. The locational attractions of the two study areas are then discussed shedding some further light on the importance of access to the pecuniary externalities available within the South East of England when set against the technological externalities available locally in Leighton Linslade and Lichfield. Finally, the relevance of technological externalities (and their institutional bases) to small firms in the two study areas is discussed in more detail.

Characteristics of small firms in Leighton Linslade and Lichfield

Table 2 provides a summary of some important characteristics of the small firms in the telephone surveys. Overall, the majority (61.6%) of small business service firms could be described as micro-firms employing five or fewer people. The vast majority (86.0%) of firms responding to the surveys were also single-site firms rather than parts of larger (usually locally-based) groups of companies. Similarly, the vast majority (83.8%) of firms in the survey were formed in the two settlements under study. A larger proportion of service firms were relocated to Lichfield than to Leighton Linslade though chi-squared tests suggested that these differences were not statistically significant. Only a minority of those firms relocating had done so from the respective nearby dominant conurbations of London and the West Midlands, the majority (three-quarters) having moved from nearby settlements such as Bedford, Hemel Hempstead, Luton and Watford in the case of Leighton Linslade and Cannock, Stafford, Tamworth and Derby in the case of Lichfield.

With all but one of the 100 firms surveyed being established since 1970, the age profile of small firms in the two case study settlements is in keeping with expectations given that the stock of small businesses in accessible rural areas tends to be markedly younger than that in remote rural and urban areas of the UK

(KEEBLE *et al.*, 1992, p. 11). Finally, overall 80.2% of firms had experienced a growth in turnover over the last three years. Overall, the proportion of firms experiencing employment growth was lower at 40.4%, though this figure conceals quite marked differences in the employment growth among businesses in the two localities; 51% of small firms in Lichfield had experienced employment growth over the previous three years compared to just 34% of firms in Leighton Linslade.

Except for their being slightly smaller and younger, the characteristics of small firms surveyed in Leighton Linslade and Lichfield are similar to those in accessible rural areas in the UK (see KEEBLE *et al.*, 1992).

Pecuniary externality fields: input and output transactions

The pecuniary effects of localized input-output linkages have been considered to lie at the heart of industrial agglomeration (e.g. SCOTT and STORPER, 1987), though recently the importance of such localized input-output linkages in contemporary instances of agglomeration and to the bulk of industry in general has been reappraised (PHELPS, 1992; STORPER, 1995). Evidence suggests that, whilst rural settlements are of some importance to small firms as markets, non-local output and input linkages will be more important to the success of small businesses in remote and accessible rural areas (KEEBLE *et al.*, 1992, p. 12). Respondents were asked to provide a breakdown of the percentage of sales made to, and inputs sourced from, various geographical areas. The results are shown in Tables 2 and 3.

From Tables 3 and 4 it is apparent that the output and input linkages of small firms in the survey are reasonably localized. On average over half of the value of output and input linkages of service firms falls within

Table 3. *Geographical patterns of sales (mean percentage)*

	Immediate locality	Conurbation and region	Rest of UK	Overseas
Leighton Linslade	12.2	58.2	22.8	7.1
Lichfield	17.5	22.1	51.1	9.7
Total	14.0	45.6	32.6	8.0
	F = 1.37	F = 53.58	F = 27.97	F = 0.39
		p < 0.01	p < 0.01	

Table 2. *Selected characteristics of small firms surveyed*

	Origin		Establishment		Date established		
	Formed	Relocated	Single-site	Group	Pre-1980	1980-89	Post 1989
Leighton Linslade	54 (84.4)	10 (15.6)	57 (89.1)	7 (10.9)	8 (12.5)	34 (53.1)	22 (34.4)
Lichfield	29 (82.9)	6 (17.1)	29 (80.6)	6 (19.5)	5 (15.6)	13 (40.6)	14 (43.8)

Note: Percentage figures in parentheses.

Table 4. Geographical source of all inputs by value (mean percentage)

	Immediate locality	Other local	Con-urbation	Rest of UK	Overseas
Leighton Linslade	22.4	34.4	11.8	30.4	0.9
Lichfield	43.1	12.7	16.1	22.9	5.2
Total	29.1	27.3	13.2	27.9	2.3
	F = 10.92		F = 13.3	F = 0.83	F = 1.05
	p < 0.01		p < 0.01		F = 2.97

the immediate locality or other local settlements. There are no significant differences in the orientation of small firms in the two locations toward local markets or overseas markets. However, an analysis of variance test reveals that there are statistically significant differences in the geography of sales between firms in the two locations with regard to value of custom derived from the nearby conurbation and surrounding region and the rest of the UK. The pecuniary effects of the supply of business service firms' products in this survey therefore operate over a wide geographical area, with the West Midlands being part of a much broader greater South East England economy extending into the Midlands. The Greater London area forms a major market for the business services industry in the UK representing a large and unique head office–business services complex (MARSHALL, 1994). It is therefore no surprise that small business service firms cite London and the South East as their main market and source of inputs – a source they can access without being physically located in it. 'Being there' may be important to the likes of business service firms (CORNISH, 1996) but it is also true to say that, for many purposes, accessibility rather than immediate location may be sufficient to effect such market presence.

Whilst Table 4 reveals some differences in the geography of input linkages of firms in the two settlements, the only statistically significant differences relate to local sourcing. Here it appears that firms in Lichfield source significantly more of their inputs (by value) from the immediate local economy and other local settlements than their counterparts in Leighton Linslade. Of note here is the established commercial infrastructure which was cited by a minority of small firms as an advantage of Lichfield (appearing within the 'other' category in Fig. 2). It is also interesting to note that the levels of local sourcing from adjacent conurbations are on average higher in the case of Lichfield than in the case of Leighton Linslade. Indeed for small firms in Leighton Linslade, other local settlements such as Milton Keynes and Luton are more important sources of inputs than London, suggesting that the pecuniary effects of mundane input-related externalities are open to small firms in accessible rural locations, since these are available over a broad area from nearby provincial towns and cities within the South East and indeed beyond.

Locational attractions of Leighton Linslade and Lichfield: access to pecuniary externalities versus local technological externalities

Respondents were asked to cite three main advantages of Leighton Linslade and Lichfield as business locations. The results are shown in Fig. 2 with the percentage of firms citing particular locational attractions on the horizontal axis and the corresponding numbers of firms at the end of each bar. Previous work has documented the importance of urban external diseconomies and rural amenity within the urban–rural shift in employment (e.g. KEEBLE, 1980; FOTHERGILL and GUDGIN, 1982) including, more recently, differential rates of small firm formation and growth (KEEBLE *et al.*, 1992; KEEBLE and TYLER, 1995). Thus, factors such as residential preference – where formation and relocation centre on the existing or preferred residence of owners or managers – and an attractive environment, on the one hand, and the availability and cost of premises, on the other, appear important attributes of Leighton Linslade and Lichfield as business locations (see Fig. 2).

Lichfield is an attractive historical city and, as one might expect, the 'pull' factors of rural amenity appear to be a particularly important aspect of Lichfield as a business location, where one-third of respondents mentioned the quality of the environment as a locational attraction. Beyond these factors composing rural amenity, both case study settlements, in and of themselves, appear to have few attractions to many small firms since the availability and cost of labour and selected other factors (including local business support infrastructure for example) are of importance to only a small proportion of firms (see Fig. 2). Further support for this point is provided in Table 5 which shows that, whilst the immediate locality is the most important source of labour employed in small firms surveyed in both case study settlements, other nearby settlements are also of importance. On the face of things these findings appear to contrast with those of Keeble *et al.* who found that labour-related factors were the second most cited contribution of accessible rural locations to the success of small firms (KEEBLE *et al.*, 1992, p. 36). However, their research sought to ascertain whether small firms in rural locations benefited from local and hinterland labour markets in terms of labour cost, turnover, etc. What both sets of findings suggest is that increases in personal mobility mean that even small firms are able to draw upon important pecuniary labour market-related externalities which operate over relatively broad geographical areas.

Furthermore, local institutions do not appear to be an important part of the attraction of Leighton Linslade and Lichfield as business locations for small firms. Respondents were asked for their opinions of the relevance of services provided by a number of local institutions and whether they currently made use of

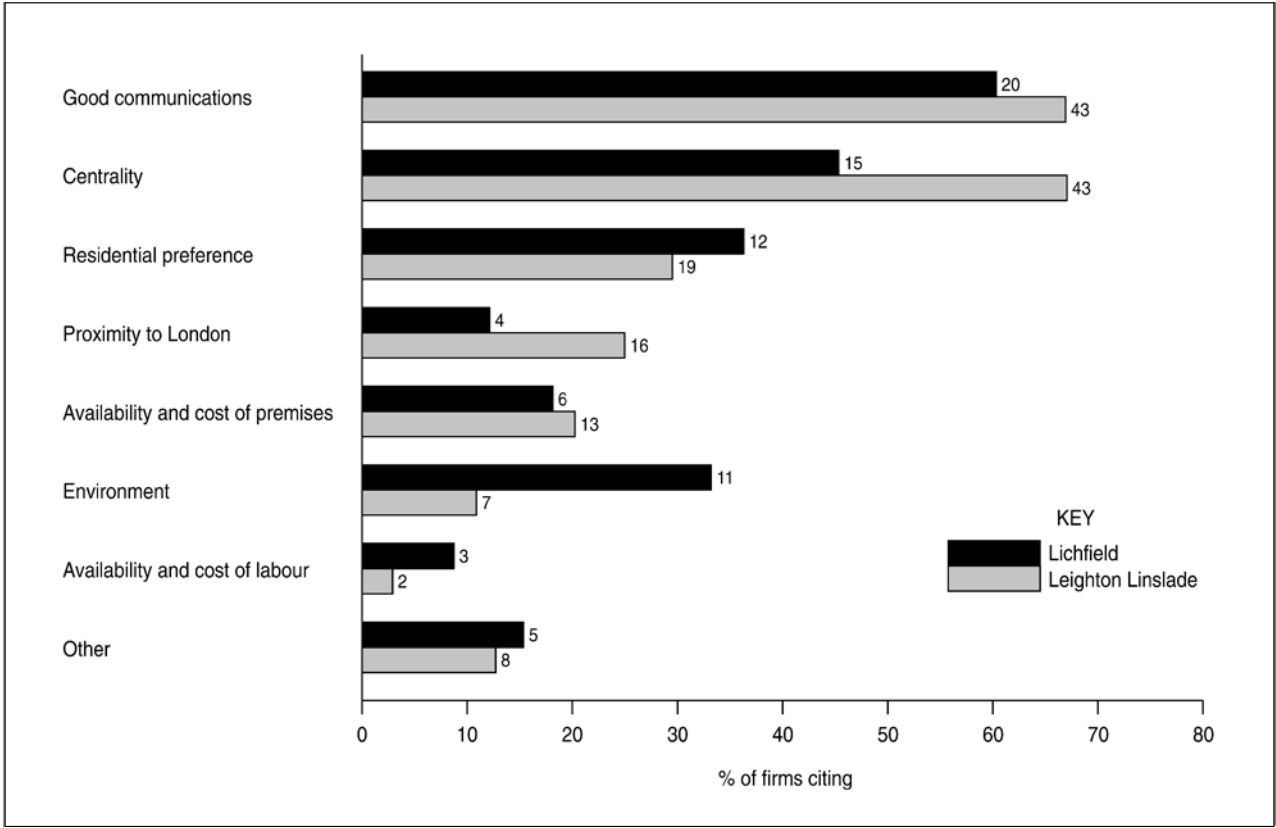


Fig. 2. Advantages of Leighton Linlade and Lichfield as business locations

Table 5. Geographical source of workforce (mean percentage)

	Immediate locality	Other local	Con-urbation	Rest of UK
Leighton Linlade	51.9	37.5	5.6	5.0
Lichfield	54.5	21.3	5.8	18.5
Total	52.9	31.6	5.6	9.9
	F = 0.94	F = 5.14 p < 0.05	F = 0.002	F = 6.82 p < 0.05

these services. The results are shown in Table 6. From this it is apparent that, with the exception of the local Chamber of Commerce and the local Business Link service in the case of Lichfield, the majority of small firms surveyed in both case study settlements do not use the services provided by local institutions. The two-by-two contingency tables produced from the final two columns in Table 6 reveal statistically significant results (at the 95% confidence limit) apart from the use of the local Chamber of Commerce. When Table 6 was collapsed into a series of two-by-two contingency tables by combining the first and second and the third and fourth columns, chi-squared tests do not reveal any statistically significant results (at the 95% confidence limit) regarding firms' perceptions of the relevance of local service providing organizations. Overall there are no important differences between the responses of small firms in the two case study settlements, rather the

results here are the product of the dominant effect of the majority of firms (irrespective of whether they are in Leighton Linlade or Lichfield) not using the services offered by local institutions and/or considering them irrelevant. The services provided by the local Training and Enterprise Council, the local authority and local colleges are considered the least relevant and those of the local Chamber of Commerce and Business Link office the most relevant to the needs of small firms, though even here a large proportion of firms consider the services irrelevant.

What Fig. 2 does show clearly is that the main attractions of both Leighton Linlade and Lichfield as business locations are in terms of their accessibility to other settlements as markets and hence sources of pecuniary external economies. Thus, good communications and centrality are the advantages of Leighton Linlade and Lichfield most often cited by small manufacturing and service firms in the survey. There are some differences between the two case study settlements in terms of the advantages cited, though these are not statistically significant (Spearman's $r=0.780$). For example, there is some support for our initial expectation that the dominant conurbations of London and the West Midlands would be an important source of technological and informational external economies to small business service firms in the case study settlements. This is particularly the case for small firms in

Table 6. *Relevance and use of selected service providers*

	Very relevant	Relevant	Not relevant	Don't know	Use	Don't use
<i>TECs</i>						
Leighton Linslade	1 (1·6)	11 (17·2)	47 (73·4)	5 (7·8)	4 (6·3)	60 (93·7)
Lichfield	2 (6·5)	8 (25·8)	16 (51·6)	5 (16·1)	8 (25·8)	23 (74·2)
<i>Local authorities</i>						
Leighton Linslade	2 (3·1)	20 (31·3)	40 (62·6)	2 (3·1)	11 (17·2)	53 (82·8)
Lichfield	4 (12·1)	4 (12·1)	21 (63·6)	4 (12·1)	8 (24·2)	25 (75·8)
<i>Business Link</i>						
Leighton Linslade	0 (0·0)	22 (34·4)	30 (46·9)	12 (18·8)	7 (10·9)	57 (89·1)
Lichfield	6 (18·2)	11 (33·3)	13 (36·4)	3 (9·0)	13 (39·4)	20 (60·6)
<i>Chamber of Commerce</i>						
Leighton Linslade	9 (14·0)	31 (48·4)	23 (35·9)	1 (1·6)	33 (51·6)	31 (48·4)
Lichfield	6 (18·7)	16 (50·0)	10 (31·3)	0 (0·0)	21 (65·6)	11 (34·4)
<i>Local colleges</i>						
Leighton Linslade	1 (1·6)	16 (25·0)	46 (71·9)	1 (1·6)	15 (23·4)	49 (76·6)
Lichfield	0 (0·0)	14 (41·2)	15 (44·1)	4 (11·8)	12 (35·3)	22 (64·7)

Note: Percentage figures in parentheses.

Leighton Linslade where proximity to London was cited by around one-fifth of small business service firms as an important attraction of Leighton Linslade as a business location. The results here do tend to contrast with the findings from Keeble *et al.*'s nationwide study where 'centrality/accessibility to clients, staff and suppliers' were of negligible importance to the success of small firms in remote and accessible rural areas (KEEBLE *et al.*, 1992, p. 36). Keeble *et al.*'s study excluded consideration of rural settlements in the South East and so our results here might indicate a contrast between the greater South East – where the phenomenon of borrowed size is important in the locational calculus of small firms – and the rest of the UK where borrowed size may be of lesser importance.

Sources of technological externalities

In the survey we sought to gain information regarding firms' sources of technical and business information in order to understand something about the locus of technological external economies accessed by small firms in the two case study settlements. The results are shown in Table 7. Overall most firms obtained their business information from national sources such as journals and exhibitions and these have been included under the 'rest of the UK' heading in Table 7. Whilst there are some differences in the main sources of business information of firms in the two settlements, a Spearman's rank correlation test indicates that these are not statistically significant.

Several observations can be made regarding the data in Table 6. First, there do appear to be some differences in the importance of the local economy to small firms in the two case study settlements. The local economy of Leighton Linslade is of almost no significance to business service firms as a source of business

information compared to Lichfield. Here we can only speculate that the importance of the immediate locality as a source of business information and competition to firms in Lichfield is a function of both its distance from Greater London and the commercial infrastructure developed during a longer history and status as a city. BRYSON and DANIELS, 1998, p. 273, have provided evidence of the manner in which small firms in the Midlands and East Anglia are partially incorporated into the 'London information nexus' in terms of the use of external business advice. Second, small firms in Lichfield appear to be linked into the regionalized economy of the South East (as seen in the proportions of firms obtaining business information from the rest of the UK).

Third, irrespective of these aforementioned differences between the orientation of small firms in the two case study settlements, the borrowing or accessing of technological external economies available in nearby large conurbations is of equal importance to small business service firms in both case study settlements. In particular, the dominant nearby conurbations of the West Midlands and Greater London appear to be an important source of business information to similar proportions of firms in both case study settlements. Small firms located in Lichfield appear to be able to borrow the benefits of size both from the regionalized economy of the South East (which exerts a more widespread effect into much of England and Wales) and, more specifically, the nearby West Midlands conurbation. Whilst the local economy in Leighton Linslade is not important as a source of business information and competition, nearby settlements such as Milton Keynes and Luton are. Settlements such as Milton Keynes and Luton are the most cited sources of market information and informal business contacts and the second most cited source of information

Table 7. *Geographical sources of business information and competition*

	Immediate local	Other local	Conurbation	Rest of UK	Overseas
<i>Market information</i>					
Lichfield	11 (32.4)	6 (17.6)	9 (26.5)	24 (26.5)	5 (14.7)
Leighton Linslade	0 (0.0)	23 (35.9)	22 (34.4)	17 (26.6)	2 (3.1)
<i>Technical information</i>					
Lichfield	5 (16.1)	6 (19.4)	5 (16.1)	26 (83.9)	6 (13.9)
Leighton Linslade	0 (0.0)	14 (21.9)	15 (23.5)	29 (45.3)	6 (9.4)
<i>Supplier information</i>					
Lichfield	16 (53.3)	8 (26.7)	8 (26.7)	16 (53.3)	4 (13.3)
Leighton Linslade	1 (1.6)	26 (40.6)	5 (7.8)	29 (45.3)	2 (3.1)
<i>Informal business contacts</i>					
Lichfield	12 (36.4)	9 (27.3)	7 (21.2)	19 (57.6)	4 (12.1)
Leighton Linslade	1 (1.6)	32 (50.0)	13 (20.3)	15 (23.4)	3 (4.7)
<i>Main competitors</i>					
Lichfield	7 (21.2)	8 (24.2)	9 (27.3)	20 (60.6)	4 (12.1)
Leighton Linslade	2 (3.1)	16 (25.0)	21 (32.8)	23 (35.9)	2 (3.1)

Note: Percentage figures in parentheses.

regarding suppliers among small business service firms in Leighton Linslade. Small firms in Leighton Linslade are therefore linked not only to the dominant Greater London economy but other important provincial urban centres elsewhere in the South East region and beyond.

CONCLUSIONS

This paper has concentrated on the relationship between small firms, the local economy and the extent to which businesses in one locality may in some way be able to borrow size from major conurbations nearby. This we suggested would be the case for a set of industries for which pecuniary externalities would be apparent at the scale of major cities or city regions. Our exploratory empirical study presented here is far from unproblematic. The UK may simply be too small an economy with too great a degree of primacy in which to examine empirically the effects of borrowed size on the location of economic activity. Here no doubt our study might usefully have considered an accessible rural settlement at a greater distance from London than Lichfield. Finally, in the paper we have explored firms' use and perception of their locality and wider business relations in simple terms, making use of descriptive and bivariate statistics to examine particular facets of individual types of external economy one by one. A further sense of the complementarity of pecuniary and technological externalities available to small firms in the sorts of accessible settlements studied here might have been gained using the sorts of multivariate statistical techniques deployed elsewhere in the literature (e.g. KEEBLE *et al.*, 1992; KEEBLE and TYLER, 1995).

To the extent that consideration of two small settlements accessible to nearby conurbations can shed light

more generally on the dynamics of small firm location in accessible rural areas, our evidence suggests that accessible rural locations may have little to offer small firms in terms of the sorts of technological externalities which are often regarded as the essence of thriving local economies. What we have tried to demonstrate and what the evidence perhaps rather imperfectly illustrates is that what these accessible locations offer is a possibly unique and complementary relationship between access to both pecuniary and technological externalities at the scale of conurbations or indeed a heavily urbanized regional economy.

In their major aggregate study KEEBLE and TYLER, 1995, have suggested that 'rural areas, and accessible rural areas in particular have economic, physical and institutional characteristics that enable enterprising behaviour to occur more readily there than elsewhere' (*ibid.*, p. 980). The physical characteristics of accessible rural areas to which Keeble and Tyler refer have not been explored explicitly in the present paper. Alongside issues of rural amenity they seem likely to be part and parcel of the way in which small firms are able to borrow the benefits without the disbenefits of urban size when locating in accessible rural locations. Yet, if indeed the phenomenon of borrowed size is apparent then the implication must be that many growth locations may have little to offer specifically in terms of technological externalities or indeed their institutional foundations. Rather they are attractive to small firms because of the access they afford to the pecuniary and technological externalities of larger nearby settlements. Viewed as an institution, the entrepreneurial labour market and personal contacts associated with it may well be important within accessible rural areas due to selective out-migration of population and businesses. However, our findings tend to suggest that the role of

formal institutional capacity (in the form of business support organizations) in accessible rural areas may easily be overstated. Future research might usefully focus on the question of whether real technological externalities or at least their institutional foundations are available to businesses in rural and accessible rural settlements, since there are clear limitations with generalizing from our own case studies but also considerable debate on this point in the literature.⁶ In stressing the relevance of borrowed size to the characteristics of small firms in smaller accessible settlements, we can suggest that it is useful to see what advantage small firms derive from not just their local economy but also the economies of nearby larger urban areas. The concept of borrowed size might therefore shed some light on the functioning of a discontinuous but nevertheless regionalized economy like the South East where many distinct settlements are in the economic orbit of Greater London (ALLEN, 1992; COE and TOWNSEND, 1998) and where there can be important relationships between provincial towns and cities (CHARLESWORTH and COCHRANE, 1994). Taken together, these observations suggest that there may be limits to what purely local economic policies may be able to achieve in terms of the development of the small firm sector. The phenomenon of borrowed size and the lack of critical mass in certain branches of industry suggest a need for policies at a regional level – involving collaboration between institutions in functionally linked settlements – and an increased relevance of national industrial policy.

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NOTES

1. To be precise it is individuals and individual firms which are able to borrow size. We take Alonso's reference to settlements being able to borrow size to be a shorthand expression of the individuals and individual businesses of given settlements, treated collectively, being embedded in similar networks of interaction through which they borrow size.
2. Indeed, despite this, pecuniary externalities are sometimes referred to as urbanization economies (e.g. KARASKA, 1969).
3. Business service industries are among the fastest growing in advanced economies and are prime examples of those presently benefiting from increases in pecuniary externalities both from general market expansion and processes of externalization (see WOOD, 1991; WOOD *et al.*, 1993).
4. Firms in the business services industries were drawn from SIC (1980) activity headings: 8370, professional and technical services n.e.s.; 8380, advertising; 8394, computer services; and 8395, miscellaneous business services.
5. This paper draws upon two essentially separate but identical studies. The first, undertaken in September 1996, looked at both manufacturing and business service firms in Leighton Lincolne as part of a student dissertation. The second, undertaken in early to mid 1998, examined Lichfield to see if borrowed size had a relevance outside the South East of England. Originally, Lichfield was chosen in order to replicate the study outside of the South East region. In the event it was possible to imperfectly replicate the study for business service firms.
6. Compare, for example, the work of KEEBLE and TYLER, 1995, which implies that local technological externalities are available and critical to the success of small firms in accessible rural areas, with that of BRYSON and DANIELS, 1998, which suggests that, in the context of an uneven knowledge economy, over-reliance on such local externalities may hinder small firm performance. These studies contrast further with CURRAN and BLACKBURN's, 1994, argument of the merely symbolic importance of the local economy to small firms.

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